

1. **Design an Applet that accepts a number from the user and displays its square when a button is clicked. Use AWT controls and appropriate event handling.**

Code:

```
import java.applet.Applet;
import java.awt.*;
import java.awt.event.*;

public class Q3_SquareApplet extends Applet implements ActionListener {
    TextField numberField;
    Button button;
    Label resultLabel;

    public void init() {
        add(new Label("Enter Number:"));
        numberField = new TextField(10);
        add(numberField);

        button = new Button("Find Square");
        add(button);

        resultLabel = new Label("Result: ");
        add(resultLabel);

        button.addActionListener(this);
    }

    public void actionPerformed(ActionEvent e) {
        try {
            double n = Double.parseDouble(numberField.getText());
            resultLabel.setText("Result: " + (n * n));
        } catch (NumberFormatException ex) {
            resultLabel.setText("Enter a valid number");
        }
    }
}
```

Output:

